Data Scientist III, Mineral Metabolism, Levi Lab

UT Southwestern Medical Center · Dallas, TX, US

Job ID (this archive)	HI-032
Employer	UT Southwestern Medical Center
Location(s)	Dallas, TX, US
Career stage (archive tag)	Experienced
Location group	DFW area
Majors (archive tags)	Biomedical Engineering, Electrical Engineering, Computer Science, Data Science
Pay listed	Not listed
Employment type	Full-time
Employer requisition	968590
Recruiting status when checked	Open when checked (Oct 7, 2026). Employer note: Readable; no closure notice found Saved PDF = full employer posting text, reformatted for print.
Source page	https://jobs.utsouthwestern.edu/job/23767006/data-scientist-iii-mineral-metabolism-levi-lab-dallas-tx/
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Course sheet refs	HI-032

Captured on 2026-10-07 from the employer's public career site. The posting text below is the employer's own wording, reformatted for printing (site navigation removed). Live listings change and close; check the source page before applying.

Job description

WHY UT SOUTHWESTERN?

With over 75 years of excellence in Dallas-Fort Worth, Texas, UT Southwestern is committed to excellence, innovation, teamwork, and compassion. As a world-renowned medical and research center, we strive to provide the best possible care, resources, and benefits for our valued employees. Ranked as the number 1 hospital in Dallas-Fort Worth according to U.S. News & World Report , we invest in you with opportunities for career growth and development to align with your future goals. Our highly competitive benefits package offers healthcare, PTO and paid holidays, on-site childcare, wage, merit increases and so much more. We invite you to be a part of the UT Southwestern team where you'll discover a culture of teamwork, professionalism, and a rewarding career!

JOB SUMMARY

Works under broad direction to provide and oversee the analysis and development of algorithms and software for clinical and/or basic research problems. Oversees and participates in programs to impact data acquisition, management, and analysis in basic, translational, and clinical research, including clinical trials.

Perform advanced professional work in support of biomedical and translational research using expertise in data science, biostatistics, computational biology, software development, and scientific visualization. Collaborate with the Principal Investigator, laboratory scientists, clinicians, and multidisciplinary partners to design rigorous analytic strategies; integrate complex biological, imaging, and clinical datasets; and deliver reproducible analyses, publication-quality figures, manuscripts, presentations, and grant materials.

This position is designed for a senior data scientist who can independently lead the analytic components of multiple research programs from study design and data acquisition through quality control, modeling, biological interpretation, validation, and dissemination. The Data Scientist III will support mechanistic and translational studies of injury, fibrosis, regeneration, aberrant mesenchymal cell fate, wound healing, tumor biology, and therapeutic response. The role is primarily computational, with close collaboration with experimental and clinical teams to ensure that analyses are biologically grounded and directly responsive to clinically important questions.

LEVI LABORATORY RESEARCH PROGRAM

The Levi Laboratory studies how immune, stromal, vascular, lymphatic, neural, epithelial, and mesenchymal cell populations interact after injury and in disease. The laboratory integrates transgenic and surgical animal models, human biospecimens, molecular and cellular biology, quantitative imaging, clinical data, and multiomic technologies to identify mechanisms, biomarkers, and therapies that promote faithful tissue repair.

Traumatic heterotopic ossification and aberrant mesenchymal differentiation after burns, musculoskeletal trauma, surgery, and tendon injury.

Fibrosis, wound healing, burn and scar biology, including inflammatory and epigenetic regulation of keratinocyte, fibroblast, immune, and progenitor-cell function.

Tendon, muscle, bone, and soft-tissue repair, including mechanotransduction, extracellular matrix remodeling, and stem or progenitor-cell fate.

Neurovascular and neuroimmune regulation of regeneration, tendinopathy, heterotopic ossification, pain, osteosarcoma, and adrenergic signaling.

Lymphatic regulation of inflammation, tissue repair, and bone resorption, including VEGF-C or VEGF-D signaling and lymphatic endothelial-cell biology.

Immunometabolism and obesity-related injury responses, including lipid metabolism, inflammatory signaling, and altered regenerative cell states.

Circulating mesenchymal progenitor cells and liquid-biopsy approaches for early diagnosis, risk prediction, and treatment monitoring.

Cancer biology and osteosarcoma, including tumor heterogeneity, tumor microenvironment interactions, angiogenesis, pain, drug response, and pulmonary metastasis.

SPECIFIC JOB DUTIES

Lead the design of data-driven research strategies with the PI, clinical collaborators, and laboratory scientists, including outcome definition, sample selection, controls, replication, power considerations, data harmonization, batch mitigation, and validation plans.

Independently analyze single-cell and bulk transcriptomic, epigenomic, spatial, proteomic, metabolomic, high-parameter cytometry, quantitative imaging, micro-CT, and clinical datasets from raw files through final biological and clinical interpretation.

Perform and document quality control, normalization, batch correction, feature engineering, cell-type annotation, differential analysis, pathway enrichment, trajectory analysis, gene-regulatory network inference, cell-cell communication modeling, and cross-modality integration.

Develop scalable image-analysis workflows for histology, immunofluorescence, multiplex imaging, confocal microscopy, whole-slide imaging, micro-CT, and three-dimensional tissue datasets, including segmentation, feature extraction, spatial quantification, and visualization.

Integrate experimental data with clinical variables, longitudinal outcomes, public datasets, and large multiomic databases to identify biomarkers, mechanisms, therapeutic targets, and clinically meaningful patient subgroups.

Develop, validate, and interpret statistical and machine-learning models for disease classification, early detection, risk prediction, treatment response, and biomarker prioritization using appropriate feature selection, cross-validation, calibration, sensitivity analyses, and external or biological validation.

Build, document, test, and maintain reproducible analysis pipelines in R and Python using version control, workflow management, containerized environments, and high-performance or cloud computing resources.

Establish and maintain laboratory standards for metadata, data dictionaries, sample tracking, file organization, quality metrics, code review, backup, controlled access, privacy, and FAIR data practices in compliance with institutional and sponsor requirements.

Partner with experimental scientists to translate computational results into specific validation studies and to ensure that data collection, assay design, and experimental metadata support rigorous downstream analysis.

Create clear, accurate, and publication-ready figures, dashboards, tables, and data narratives for manuscripts, grants, abstracts, presentations, progress reports, and institutional reviews.

Lead computational and quantitative sections of manuscripts and grant applications, contribute to methods and results writing, maintain transparent analytic documentation, and support responses to scientific and peer review.

Present findings at laboratory meetings and national or international scientific forums, communicate limitations and uncertainty clearly, and provide analytically sound recommendations to research teams.

Mentor trainees and staff in study design, statistics, computational biology, reproducible research, scientific visualization, and responsible interpretation of complex data.

Manage multiple concurrent projects, establish priorities and milestones, identify analytic risks early, and move work efficiently from exploratory analysis to validated, publication-quality completion.

PREFERRED TECHNICAL EXPERTISE

Ph.D. in data science, biostatistics, bioinformatics, computational biology, computer science, biomedical engineering, or a related quantitative or biological field, with substantial experience applying advanced analytics to biomedical research.

Master's degree in a related quantitative or biological field and significant relevant experience independently leading complex biomedical data analyses may be considered.

Bachelor's degree in a related quantitative or biological field with extensive progressively responsible experience in biomedical data science may be considered.

Advanced proficiency in R and Python for data analysis, statistics, visualization, and pipeline development, with experience using Linux, shell scripting, Git, high-performance computing, and structured collaborative code review.

Demonstrated ability to independently analyze at least two major classes of complex biomedical data, such as bulk or single-cell sequencing, spatial omics, quantitative imaging, clinical outcomes, proteomics, metabolomics, or high-parameter cytometry.

Strong command of statistical inference, experimental design, regression and generalized models, multiple-testing correction, missing-data methods, confounding, batch effects, predictive-model evaluation, and the distinction between exploratory and confirmatory analyses.

Ability to translate biological and clinical questions into rigorous analytic plans, explain technical concepts to nontechnical collaborators, and communicate uncertainty without overinterpreting results.

Demonstrated ability to evaluate emerging computational methods, benchmark their performance, and implement appropriate tools with transparent documentation, testing, and quality controls.

Candidates should provide a complete publication record, such as ORCID, Google Scholar, or PubMed, and a link to a GitHub profile, code repository, or other portfolio demonstrating reproducible analytic work.

Experience with single-cell or spatial transcriptomics and at least one additional high-dimensional modality such as metabolomics, proteomics, epigenomics, CyTOF, imaging mass cytometry, multiplex tissue imaging, or quantitative micro-CT.

Experience managing and integrating large clinical, imaging, or multiomic databases with robust metadata, reproducible provenance, privacy protections, and appropriate access controls.

Experience developing interpretable machine-learning or statistical-learning models and evaluating performance, calibration, transportability, bias, and biological or clinical validity.

A record of peer-reviewed publications and documented analysis code or research software demonstrating meaningful contributions to computational, translational, or biomedical research.

PREFERRED DOMAIN EXPERIENCE

Mesenchymal progenitor and stem-cell biology, lineage tracing, aberrant cell fate, osteogenesis, chondrogenesis, fibrosis, wound repair, or extracellular matrix biology.

Immune-stromal, neurovascular, lymphatic, or tumor microenvironment analysis, including macrophage states, endothelial populations, sensory and sympathetic nerves, and intercellular signaling.

Preclinical models of burn and musculoskeletal injury, heterotopic ossification, tendon degeneration, obesity or metabolic disease, wound healing, cancer, or metastasis.

Human translational studies using biospecimens, circulating rare-cell populations, biomarker discovery, diagnostic modeling, imaging, electronic health record data, or longitudinal clinical outcomes.

SCIENTIFIC LEADERSHIP AND PROFESSIONAL COMPETENCIES

Demonstrated ownership of complex analyses, sound scientific judgment, careful documentation, timely delivery, and the ability to prioritize effectively across multiple collaborative projects.

Excellent critical thinking, troubleshooting, scientific writing, oral communication, collaboration, and project-management skills.

Ability to provide constructive analytic leadership, mentor colleagues, improve laboratory data practices, and build productive partnerships among computational, experimental, and clinical teams.

Commitment to research integrity, reproducibility, responsible authorship, data stewardship, privacy, continuous learning, and sustained engagement with the scientific questions underlying each analysis rather than functioning solely as a data-processing

service.

BENEFITS

UT Southwestern is proud to offer a competitive and comprehensive benefits package to eligible employees. Our benefits are designed to support your overall wellbeing, and include:

PPO medical plan, available day one at no cost for full-time employee-only coverage

100% coverage for preventive healthcare-no copay

Paid Time Off, available day one

Retirement Programs through the Teacher Retirement System of Texas (TRS)

Paid Parental Leave Benefit

Wellness programs

Tuition Reimbursement

Public Service Loan Forgiveness (PSLF) Qualified Employer

Learn more about these and other UTSW employee benefits!

EXPERIENCE AND EDUCATION

Required

Education

PhD In Computer Science, Bioinformatics, (Bio)mathematics, (Bio)statistics, Physics, Electrical Engineering, or related field or

Master's Degree M.S. in Computer Science, Bioinformatics, (Bio)mathematics, (Bio)statistics, Physics, Electrical Engineering, or related field or

Bachelor's Degree B.S. with demonstrated experience in broadly defined areas of bioinformatics

Experience

2 years of post-graduation experience in data analysis and/or scientific software development with PhD

4 years of post-graduation experience in data analysis and/or scientific software development with Master's degree

6 years of post-graduation experience in data analysis and/or scientific software development with Bachelor's degree

JOB DUTIES

Analyzes and develops formalisms, algorithms and software for applications in biomedical research.

Develops and maintains software tools and infrastructure for resolving specific problems.

Oversees the development, testing and revision of programs in a team with diverse bioinformatics expertise.

Trains users and other personnel in analyzing, defining and resolving computational problems.

Performs other duties as assigned.

SECURITY AND EEO STATEMENT

Security

This position is security-sensitive and subject to Texas Education Code 51.215, which authorizes UT Southwestern to obtain criminal history record information.

EEO

UT Southwestern Medical Center is committed to an educational and working environment that provides equal opportunity to all members of the University community. As an equal opportunity employer, UT Southwestern prohibits unlawful discrimination, including discrimination on the basis of race, color, religion, national origin, sex, sexual orientation, gender identity, gender expression, age, disability, genetic information, citizenship status, or veteran status.

Notes for ECS 1100 students

How this connects to your major

Data Science: healthcare data analysis, data engineering, statistics, AI or computational biology. These are teaching annotations, not an employer guarantee of eligibility.

Use the degree and experience evidence to distinguish a first job from a later career step.

DS: Named discipline appears in source; consult qualifications for eligibility

Skills mentioned in the posting

Python, R, Git/source control, Statistics, Data visualization, Cloud, Verification/validation, Clinical knowledge, Genomics/bioinformatics, Communication/teamwork

Eligibility excerpts not shown above

Compensation (source excerpts): Not stated in captured posting text.

Course sheet tags

Majors (teaching tags): DS; Career stage (teaching tag): Senior / leadership; Geography: DFW; Recruiting status at capture: Posting readable at capture; no closure notice found; Employment type: Full-time.

Source record

Original posting https://jobs.utsouthwestern.edu/job/23767006/data-scientist-iii-mineral-metabolism-levi-lab-dallas-tx/

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Saved raw source: sources/postings/utsw_23767006.json

SHA-256: 18d24e82509a01d4efee15556e623253c1e3dea81cffd3ae1b7d6699aa223b31